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A *Carex* species-dominated marsh community represents the best short-term target for reclaiming wet meadow habitat following oil sands mining in Alberta, Canada

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ABSTRACT

Oil sands mining creates vast areas of disturbed land with contaminated sediment and sub-saline water that require reclamation. Thus far reclamation of wetlands on oil sand leases has been sporadic, although some marshes and shallow water wetlands developing on the leases have begun to exhibit viable vegetation communities. To identify the communities found on oil sands leases that demonstrate ecological performance similar to that of natural wetlands, and to provide guidance on specific revegetation practices, we compared the vegetation communities of 25 natural fresh to sub-saline marshes, which represent the most realistic outcome of the reclaimed wetlands, and 20 oil sands reclaimed marshes. We found that wetlands did not have consistent vegetation communities on the basis of their construction history, and that exposure to oil sands process-affected water does not determine the vegetation community that will develop in reclaimed oil sands wetlands. We identified three wet meadow vegetation communities among our study wetlands using the statistical technique of hierarchical cluster analysis: sedge-dominated natural reference, reclaimed sedge, and disturbed/saline. Notably, the reclaimed sedge communities in the oil sands sites were different from sedge communities in natural marshes; *Carex aquatilis* Wahlenb. dominates reclaimed sites whereas *Carex atherodes* Spreng. dominates natural sites. The majority of vegetation communities that develop in reclaimed wetlands produce less aboveground biomass than natural wetlands, despite having similar species richness and percent cover. In general, oil sands reclaimed wetlands have lower levels of sediment nutrients and lower sediment water content than natural wetlands, and these deficiencies may be limiting vegetation community biomass production compared to natural wetlands. The presence of a *Carex* species-dominated vegetation community at some reclaimed sites shows promise for future reclamation success, if revegetation targets the establishment of this community.

1. Introduction

Alberta, Canada, holds the world's third largest reserve of crude oil, 99% of which is found in oil sands deposits that are currently being extracted by surface and sub-surface mining (ERCB, 2009). The total surface mineable oil sands area in Alberta is 4800 km^2, with 1670 km^2 either being actively mined or approved for development within the next decade (Government of Alberta, 2012; Rooney et al., 2012). Already 715 km^2 of boreal landscape have been destroyed to access this resource (Government of Alberta, 2012), creating open pits and expanses of tailings, with toxic elements in water and sediment such as naphthenic acids (NAs), metals, polycyclic aromatic hydrocarbons (PAHs), elevated salinity, as well as unstable and nutrient poor substrates that impede plant colonization (Giesy et al., 2010; Rooney and Bayley, 2011).

Provincial legislation requires oil sands companies to reclaim the post-mining landscape to "equivalent land capability" (Harris, 2007), to support land uses after reclamation that approximate those that existed before mining (Alberta Environment, 1999). It is unlikely, and unrealistic to expect that oil sands wetlands will return to the same peatlands that existed on the pre-mining landscape (Johnson and Miyanishi, 2008; Purdy et al., 2005; Rooney et al., 2012), which were greater than 60% wetland by area, the majority being fen or bog peatlands (Rooney et al., 2012). In fact, wetlands are expected to comprise only one quarter of the reclaimed landscape (Rooney et al., 2012). Given the mineral substrate and elevated sediment and water salinities of the

Abbreviations: REF, reference treatment; OSPA, oil sands process-affected treatment; OSREF, oil sands reference treatment; WM, wet meadow; EM, emergent; OW, open water; NMS, non-metric multidimensional scaling; IBI, Index of Biotic Integrity; NA, naphthenic acid; PAH, polycyclic aromatic hydrocarbon.

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post-mining landscape, construction of shallow, sub-saline open water and marsh wetlands may represent the best endpoint for reclaimed wetlands (Purdy et al., 2005), and while natural analogs of such wetlands exist, they are infrequent in the boreal plain of northern Alberta (Trites and Bayley, 2009a). To ensure equivalent land capability is reached it is critical that wetland plants be established on the mined landscape (Harris, 2007; Johnson and Miyanishi, 2008).

Marshes and shallow open water wetlands have already begun to form on oil sands leases with some constructed for research purposes, while others have developed unintentionally in areas of suitable substrate and hydrology (Trites and Bayley, 2009a). A few wetlands appear to have developed productive marsh vegetation (Trites and Bayley, 2009a,b), contain native wetland plant species that tolerate contaminants and salinity at the levels found in the post-mining oil sands landscape (Mollard et al., 2012; Rezanezhad et al., 2012; Vitt et al., 2011), and have ecological conditions within the range of natural wetlands, as measured by a vegetation-based Index of Biotic Integrity (IBI; Raab and Bayley, 2012). Yet the wetlands that demonstrate good ecological conditions do not share a common construction history, and vary across a range of contamination (Raab and Bayley, 2012).

The success of reaching equivalent land capability relative to analogous natural vegetation communities can be measured using individual indicators of performance such as percent cover and peak aboveground biomass, or indices such as the vegetation-based IBI that integrate a number of these individual indicators. Indices of Biotic Integrity have been used extensively in other jurisdictions to assess the relative health of reclaimed wetlands (e.g., Balcombe et al., 2005; Gutrich et al., 2009), however a common criticism on the use of IBIs is that the loss of information when many variables are collapsed into a single index could mask real differences between groups of sites (McCoy and Mushinsky, 2002), and therefore lose information that may be useful in informing reclamation practices.

Another well-established method to assess wetland condition while retaining information on the plant community is to examine differences in species composition among natural and reclaimed wetlands. Community composition can be determined using multivariate statistical techniques to highlight differences among wetland types (Balcombe et al., 2005; Collier, 2009), by grouping sites into communities based on the presence of dominant species, and are useful in detecting trends in species as well as physical and chemical variables among sites (Keleher and Rader, 2008; McCune and Grace, 2002).

By delineating distinct marsh vegetation communities among reclaimed and natural boreal marshes, then comparing their performance using the criteria of vegetation percent cover (to stabilize sediment and provide habitat), peak aboveground biomass (to contribute to sediment organic matter and enter the food chain), and similarity of species composition to natural analogs we can identify which communities on oil sands leases show promise for meeting the level of ecological performance demonstrated by natural wetlands, and identify which communities and species are best targeted for reclamation. The objectives of this study were to (1) sample natural reference wetlands and oil sands reclaimed wetlands across a range of physical and chemical conditions, and group sites on the basis of their plant communities, (2) determine how vegetation communities on reclaimed oil sands marshes compare to communities present at natural sites, (3) assess how each vegetation community performs ecologically in the context of reaching equivalent land capability, and (4) identify specific communities and/or species that may be targeted for reclamation on oil sands leases. In addition, we attempt to make inferences as to the physical and chemical factors, including current reclamation practices that can impede the colonization and successful development of a viable vegetation community.

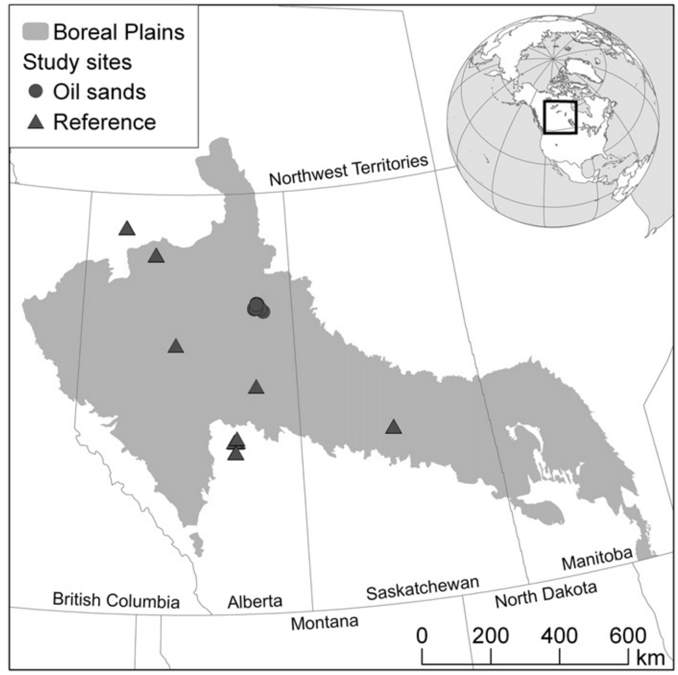

Fig. 1. Study site locations ($n=45$) within the boreal plains ecoregion of northern Alberta and Saskatchewan. Circle = oil sands reclaimed (OSREF, OSPA) site, triangle = natural reference (REF) site. Sites located to the south of the zone in central Alberta are found within a small pocket of boreal forest, discontinuous with the rest of the zone. Sites to the north of the boreal plain in northern Alberta are within a zone of transition to the taiga plains. Not all sites are visible due to symbol overlap.

2. Methods

2.1. Study sites

Reclaimed oil sands and natural reference sites were visited in 2007 and 2008 within the North American boreal plains ecoregion (Fig. 1, Table A.1). The boreal plains are characterized by flat topography with surficial glacial deposits of loamy till and gravel-sand glaciofluvial 30–200 m deep overlying Mesozoic- and Cenozoic-age sedimentary bedrock (Johnson and Miyanishi, 2008). The subhumid mid to high boreal climate annually averages −2 to +1 °C, with 500–700 mm of precipitation annually, 70% of which occurs between May and September (Devito et al., 2000). Low topography and deep glacial deposits in this region result in complex groundwater patterns that are poorly documented (Price et al., 2005; Zoltai et al., 1988). The pre-development landscape is composed of upland forest, peatlands, open water wetlands, and lakes. The wetlands of the boreal plain are primarily peat-producing fens and bogs, although small areas of marshes and shallow open water wetlands (ranging in salinity from fresh to sub-saline) can also be found (Rooney et al., 2012; Trites and Bayley, 2009a). Forty-five wetlands were selected based on their construction history (which we call treatments): 20 reclaimed wetlands on oil sands leases, and 25 natural reference wetlands located away from anthropogenic disturbance, usually in protected areas.

The oil sands leases of Syncrude Canada Limited and Suncor Energy Incorporated near Fort McMurray, Alberta, Canada (56.8531°N, 111.3180°W to 57.1150°N, 111.6833°W), where our reclaimed sites were located, contain the oldest and most extensive oil sands reclaimed wetlands. These are predominately fresh to sub-saline shallow open water with marsh fringes that have been both intentionally constructed, or formed "opportunistically" where conditions on the post-mining landscape have allowed. With the exception of the single youngest site, all 20 reclaimed sites were seven years of age or older at the time of sampling. Ages ranged from 3 to 35 years, with an average age of 16 years. We classified oil sands reclaimed wetlands a priori into two treatments, on the basis of their construction and physicochemical history. We considered whether a wetland was subjected to solely physical disturbance such as gravel extraction or impoundment (oil sands reference treatment [OSREF]), or physical disturbance and chemical contamination (oil sands process-affected treatment [OSPA]) through exposure to oil sands process water or substrates such as tailings that are often highly saline, and can contain NAs, PAHs as well as other contaminants. This disturbance could have occurred in a single period in the history of the wetland, or is ongoing, as in the case of some wetlands receiving seepage water from nearby tailings facilities. OSREF sites can also have increased salinity due to the weathering of marine shale overburden excavated to the surface.

Natural reference (REF) wetlands, our third treatment type, were located at six loci across the boreal plain (53.6015°N, 105.8957°W to 59.1098°N, 118.0797°W). Natural reference wetlands were shallow open water marshes, and spanned a range in salinity from fresh to sub-saline. The natural rarity of REF sites that met our criteria necessitated the large spatial spread of sites. The age of natural wetlands was unknown, although it is estimated that they have been on the landscape for >1000 years, and may have experienced periodic alteration by beaver (*Castor canadensis* Kuhl) activity.

We captured similar size ranges among REF and OSREF treatments (0.6–24 ha and 0.8–25 ha, respectively), however, OSPA wetlands were generally smaller in size (0.4–3.2 ha). Mean wetland size for all treatments was 4.5 ± 5.6 ha.

Three wetland vegetation zones occurred at our study wetlands: the wet meadow, emergent, and the shallow open water zone (Raab and Bayley, 2012). The wet meadow (WM) zone contains sedges (*Carex* spp. L.), grasses, and wetland forb species, and often is inundated in spring until the water level drops below the sediment surface for the majority of the growing season. The emergent (EM) zone occurs between the wet meadow and open water, contains robust wetland-obligate emergent species, and is typically flooded for the duration of the growing season.

2.2. Vegetation sampling

Vegetation data were collected in August at growing season peak biomass using methods described in Rooney et al. (2011). Briefly, six 1 m^2 plots in each zone (WM and EM) at each wetland were assessed for species composition and percent cover by species. A power analysis of preliminary species and cover data from 2007 using PASS v6.0 (NCSS, 1997) indicated that six plots was sufficient to capture among-site difference at our study sites. When it was not possible to identify plants to the species level, genus was used, and nomenclature followed the USDA PLANTS Database (USDA, NRCS, 2012). To obtain a measure of aboveground biomass production, macrophyte aboveground growth in 0.25 m^2 biomass plots was clipped to within 1 cm of the substrate surface and sorted by species or in a few instances genus, dried to constant mass and weighed.

2.3. Physicochemical sampling

Physical and chemical data were collected in August of 2007 and 2008, as described in Rooney et al. (2011). Data included water and sediment chemistry, as well as physical variables (e.g., water depth). Rooney and Bayley (2010) described the local abiotic conditions at study sites by integrating the level of physical and chemical stress occurring at each location, and created an abiotic gradient of stress that chose eight variables from an overall set of 52. The stress score of each site was calculated from: open water (OW) cations, OW total nitrogen, OW chloride, EM zone sediment % water and EM zone sediment oil content, maximum OW depth, OW secchi depth as a proportion of total depth, and OW amplitude over the growing season. The stress gradient incorporated elements of water chemistry, sediment chemistry, and wetland morphology, and was not intended to distinguish between anthropogenic and natural disturbance, thus we used this approach to include an objective measure of abiotic stress at study sites.

2.4. Data screening

For each vegetation zone (WM and EM) at each wetland, six 1 m^2 vegetation plots were combined to provide species composition values representative of the entire zone. Rare species were removed from community composition data before multivariate data analyses, to reduce noise and enhance the likelihood of detecting relationships among species communities and environmental variables (McCune and Grace, 2002). Species were deleted if they were present at less than 5% of sites. Percent cover data, as proportions, were arcsin square root transformed before multivariate analyses (Sokal and Rohlf, 1995).

To assess whether the WM and EM vegetation communities at our sites were independent of each other, we performed a Mantel Test with the randomization (Monte Carlo) method of interpretation and Bray–Curtis distance measure (Sokal and Rohlf, 1995). The results indicate non-independence of the WM and EM zone species composition data (r-statistic = 0.314, $p < 0.001$). Therefore we considered the two zones to be redundant in our analyses, and all subsequent analyses and results are based on only WM data, which tended to be more species-rich.

2.5. Approach

We chose to group sites on the basis of their vegetation community composition rather than treatment (i.e., REF, OSREF, OSPA), which has been found to be a poor predictor of wetland vegetation at oil sands reclaimed wetlands (Raab and Bayley, 2012; Trites and Bayley, 2009a). Multivariate techniques facilitate comparison among study sites on the basis of their species composition (McCune and Grace, 2002), and can be particularly effective when assessing community similarity or looking for species and physicochemical trends among highly variable or degraded sites (Collier, 2009; Keleher and Rader, 2008; Reynoldson et al., 1997). We used hierarchical cluster analysis to group sites, identified the species that characterized these vegetation communities using Indicator Species Analysis, and examined whether there were significant differences in ecological performance among the vegetation communities we found by comparing aboveground biomass and vegetation cover. Finally we visually explored species and physicochemical trends among communities plotted in species space using non-metric multidimensional scaling (NMS), to elucidate potential physical or chemical barriers to reclaiming healthy vegetation communities.

2.5.1. Site grouping and communities

To group sites by vegetation community we used hierarchical cluster analysis (distance measure: Bray–Curtis, linkage type: flexible beta −0.25) on community composition data. To determine the optimal number of groups we used Indicator Species

Table 1
Mean and total species at wetland treatment types and hierarchical cluster analysis-determined communities. Treatments: REF = natural reference, OSREF = oil sands reference, OSPA = oil sands process affected.

	n	Mean (±SE) species per site	Total species per community/treatment
Treatment			
REF	25	24.9 (1.51)	120
OSREF	9	25.3 (2.24)	85
OSPA	11	23.8 (2.57)	85
Community			
Natural sedge (natsedge)	21	25.8 (1.18)	112
Reclaimed sedge (recsedge)	13	21.4 (2.67)	86
Disturbed/saline (distsal)	11	26.6 (2.24)	103

Analysis (ISA) with a Monte Carlo test of significance (Dufrene and Legendre, 1997). The optimal group number was selected on the criteria of low average Monte Carlo test p-value, and higher numbers of significant indicators. We used ISA to identify the characteristic, or indicator species of each community type. All multivariate analyses were carried out using PC-ORD v4 (McCune and Grace, 2002; McCune and Mefford, 1999).

2.5.2. Ecological performance among communities

To assess variation in aboveground biomass production and vegetation percent cover among vegetation communities we used one-way ANOVA with Tukey's multiple comparisons in SPSS v17 (SPSS Inc., 2008). Data were square root transformed when assumptions of heterogeneity were not met.

2.5.3. Vegetation and environmental patterns

To observe the grouping of sites in multivariate species space, and to examine major species composition and environmental gradients, we performed NMS using the Bray–Curtis distance measure in PC-ORD v4 (McCune and Grace, 2002; McCune and Mefford, 1999). To determine the optimal dimensionality for the NMS we used an NMS scree plot and then ran the model with this dimensionality from a random starting configuration.

To examine patterns in species composition and underlying physical and chemical gradients among study sites, species and physicochemical variable overlays were displayed on NMS plots. All species used in the vegetation community analysis matrix were included as potential joint plot vectors (Table A.2). The eight potential physicochemical joint plot variables were WM sediment C, N, P, and water content, water depth in vegetation plots, OW conductivity (salinity), OW NAs, and the abiotic disturbance score of each site. Overlay vectors were plotted only if they had correlations with the NMS of $R^2 > 0.20$ in order to explore trends, though not necessarily infer statistical significance.

3. Results

We detected 134 species during vegetation surveys in August of 2007 and 2008 (Table A.2). Mean per-site species richness among REF, OSREF, and OSPA treatments was similar, while total species richness by treatment was highest for REF sites, with 120 species, and equal for both OSREF and OSPA, with 85 species each (Table 1). Data screening to strengthen subsequent multivariate analyses reduced the overall species number by greater than 50%, from 134 to 62.

3.1. Site grouping and communities

The 45 sites clustered optimally into three vegetation communities (hierarchical cluster analysis and ISA; 20% information remaining, 2.17% chaining). On the basis of their component vegetation and physical locations these communities were titled the natural sedge (natsedge) community, reclaimed sedge (recsedge) community, and disturbed/saline (distsal) community (Fig. 2; Table 2). The majority (>75%) of REF sites were defined by the natsedge community, indicated by the presence of *Carex atherodes* Spreng. (swallow-tail sedge), and *Scutellaria galericulata* L. (marsh skullcap). Two OSREF sites on oil sands leases also supported the natsedge vegetation community, both containing *C. atherodes*.

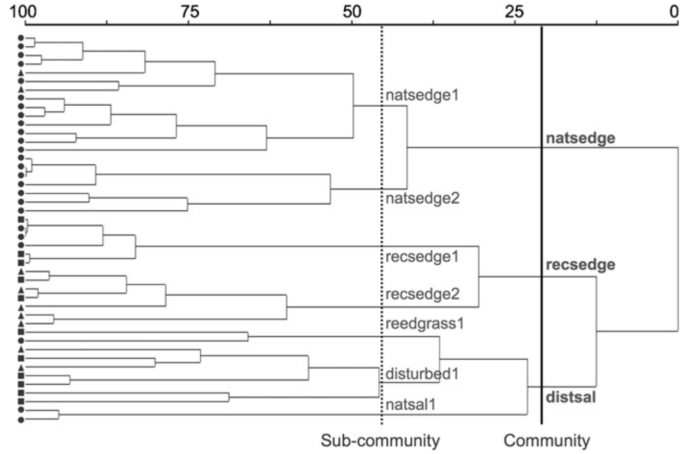

Fig. 2. Marsh communities and sub-communities from cluster analysis dendrogram (distance measure: Bray–Curtis, linkage type: flexible beta −0.25) from arcsin square root transformed percent cover data, with percent information remaining indicated along the top. Wetland treatment type is indicated by symbols on the left: REF (circle), OSREF (triangle), and OSPA (square). Vertical lines indicate pruning location at the community (solid line) and sub-community (dashed line) levels, with each community and sub-community denoted: natsedge = natural sedge, recsedge = reclaimed sedge, distsal = disturbed/saline, natsedge1 = natural sedge I, natsedge2 = natural sedge II, recsedge1 = reclaimed sedge I, recsedge2 = reclaimed sedge II, reedgrass1 = Reedgrass, disturbed1 = disturbed, natsal1 = natural saline.

The recsedge community was found at an equal number of OSPA and OSREF sites, however three REF sites also supported the recsedge community. While still sedge-dominated like the natsedge community, the dominant sedge species in the recsedge community differed with these sites primarily characterized by *C. aquatilis*, which was the significant indicator species of this community (Table 2).

The distsal community was the only community not dominated by sedge species, and was found at primarily OSPA sites, but did occur at two OSREF sites, and three REF sites. The significant indicator species of this community were *Hordeum jubatum* L. (foxtail barley, a salinity-tolerant grass species), and *Sonchus* spp. L. (sow thistle), which is considered noxious in Alberta (Alberta Weed Control Act, 2010).

Due to the unexpected inclusion of REF sites in the disturbed/saline community group, we chose to examine the three communities in greater detail by sub-dividing them into sub-communities. Returning to the hierarchical cluster analysis, Indicator Species Analysis and Monte Carlo p-values identified that further division of the three vegetation communities into seven vegetation sub-communities (at 45% information remaining) was the next most obvious pruning level (Fig. 2; Table 3). We found both the natsedge and recsedge communities each were composed of two sub-communities. The distsal community divided into three sub-communities: Reedgrass (reedgrass1), disturbed (disturbed1), and natural saline (natsal1).

The natsal1 sub-community describes the vegetation of two REF sites, and seven species of halophytic vegetation were associated with the natsal1 sub-community as significant indicators

Table 2
Results of Indicator Species Analysis at the community group level. Indicator species shown had ISA maximum indicator value > 0.5, $p < 0.01$. Treatments: REF = natural reference, OSREF = oil sands reference, OSPA = oil sands process affected.

Community	Indicator species	*n* (treatment membership)
Natural sedge (natsedge)	*Carex atherodes* Spreng., *Scutellaria galericulata* L.	21 (19 REF, 2 OSREF)
Reclaimed sedge (recsedge)	*Carex aquatilis* Wahlenb.	13 (3 REF, 5 OSREF, 5 OSPA)
Disturbed/saline (distsal)	*Hordeum jubatum* L., *Sonchus* spp. L.	11 (3 REF, 2 OSREF, 6 OSPA)

(Table 3). The disturbed1 sub-community was characterized by *Melilotus* spp. Mill., (sweetclover), and was found at five OSPA and two OSREF wetlands. The reedgrass1 sub-community, indicated by *Calamagrostis stricta* (Timm) Koeler., was found at two sites, one REF wetland and one OSREF wetland.

The age of reclaimed sites with the recsedge community was not significantly different from reclaimed sites with the distsal community (17 ± 2.9 years and 15 ± 4.3 years, respectively; One-way ANOVA; $F(1, 16) = 0.063$, $p = 0.81$); that is, age was not a determining factor in the vegetation communities we observed at oil sands sites. The average age of sites with the natsedge vegetation community was assumed to be > 1000 years, as these were primarily REF sites.

3.2. Ecological performance among communities

The natsedge community produced significantly higher August peak aboveground biomass than either the recsedge or distsal communities (One-way ANOVA with Tukey's multiple comparisons; $F(2, 42) = 17.42$, $p < 0.001$), with nearly 50% greater peak aboveground biomass (Fig. 3A). The majority of aboveground biomass was attributed to *Carex* spp. in the natsedge and recsedge communities, whereas *Carex* spp. comprised less than 20% of the aboveground biomass in the distsal community (Fig. 3A). In contrast to the results for total aboveground biomass, *Carex* spp. aboveground biomass was not significantly different among the natsedge and recsedge communities, both of which had higher *Carex* spp. biomass than the distsal community (One-Way ANOVA with Tukey's multiple comparisons; $F(2, 42) = 48.66$, $p < 0.001$). Limitations in the field prevented the separation of aboveground biomass contributions below the genus level.

Percent cover of vegetation in the natsedge and recsedge communities were similar, despite differences in aboveground biomass (Fig. 3B). The distsal community had significantly lower vegetation cover than the natsedge community, or inversely, a significantly greater area of bare ground (One-way ANOVA with Tukey's multiple comparisons; $F(2, 42) = 6.26$, $p < 0.01$).

Only within the distsal community were there significant differences among sub-community aboveground biomass and percent cover. Aboveground biomass in the natsal1 sub-community was significantly greater than the disturbed1 sub-community (Fig. 3C; One-way ANOVA with Tukey's multiple comparisons; $F(2, 8) = 11.86$, $p < 0.01$). Vegetation percent cover in the natsal1 sub-community was also significantly greater than the disturbed1 sub-community (Fig. 3D; One-way ANOVA with Tukey's multiple comparisons; $F(2, 8) = 19.26$, $p < 0.01$). The reedgrass1 sub-community, however, did not differ significantly from either of the two other distsal sub-communities, as the high variability of these values combined with low sample sizes limited the detection of statistically significant differences.

3.3. Vegetation and environmental patterns among communities

An NMS scree plot identified two axes as the best trade-off between the amount of information displayed on a single plot and stress in the model. The NMS was run with this dimensionality, and reached low final instability (<0.00001) after 112 of 200 possible iterations. The final stress of the model was near the upper limit of what is considered biologically interpretable (final stress = 19.80), however large sample sizes, such as our 45 sites, are known to increase NMS stress while still providing interpretable results (McCune and Grace, 2002). Both Axis 1 and 2 were significant (Monte-Carlo $p < 0.05$) and together represented 74% of the variance in the model. The three vegetation communities identified by hierarchical cluster analysis formed visually distinct clusters in an NMS plot (Fig. 4).

Species overlay vectors indicated that Axis 1 was primarily a *C. atherodes* – *Equisetum* spp. L./*Sonchus* spp. gradient (Fig. 4 inset A). This axis separated the natsedge community from both the recsedge and distsal communities. Higher cover of *C. atherodes* was found at left on the axis where the natsedge community sites were grouped, while *Equisetum* spp. (horsetail) and the noxious *Sonchus* spp. increased to the right on the axis. Axis 2 was a *C. aquatilis* – halophyte species (*H. jubatum*, *Glaux maritima* L., *Scolochloa festucacea* (Willd.) Link., *Puccinellia nuttalliana* (Schult.) Hitchc., *Elymus trachycaulus* (Link) Gould ex Shinners) gradient, and separated the recsedge community from the distsal community. *C. aquatilis* was found high on Axis 2, where the recsedge community was clustered.

With respect to a priori treatment type, REF wetlands were clustered to the left on Axis 1, while both OSPA and OSREF treatment wetlands were distributed to the right on Axis 1 and evenly along Axis 2 indicating that the variation in vegetation communities among these wetlands was due to changes in percent cover of *C. aquatilis* and halophyte species (in particular *H. jubatum*).

Table 3
Results of Indicator Species Analysis at the sub-community group level. Indicator species shown had ISA maximum indicator value > 0.5, $^{**}p < 0.01$, $^{*}p < 0.05$. Treatments: REF = natural reference, OSREF = oil sands reference, OSPA = oil sands process affected.

Sub-community	Indicator species	*n* (treatment membership)
Natural sedge I (natsedge1)	*Scutellaria galericulata* L.**	14 (12 REF, 2 OSREF)
Natural sedge II (natsedge2)	*Carex atherodes* Spreng.**	7 (7 REF)
Reclaimed sedge I (recsedge1)	*Carex aquatilis* Wahlenb.**	6 (3 REF, 3 OSPA)
Reclaimed sedge II (recsedge2)	*Carex utriculata* Boott**	7 (5 OSREF, 2 OSPA)
Reedgrass (reedgrass1)	*Calamagrostis stricta* (Timm) Koeler**, *Achillea sibirica* Ledeb.**, *Achillea millefolium* L.**, *Typha latifolia* L.*	2 (1 REF, 1 OSPA)
Disturbed (disturbed1)	*Melilotus* spp. Mill.*	7 (2 OSREF, 5 OSPA)
Natural saline (natsal1)	*Glaux maritima* L.**, *Hordeum jubatum* L.**, *Puccinellia nuttalliana (Schult.) Hitchc.***, *Schoenoplectus maritimus* (L.) Lye **, *Elymus trachycaulus* (Link) Gould ex Shinners **, *Scolochloa festucacea* (*Willd.*) *Link**, *Triglochin maritima* L.*	2 (2 REF)

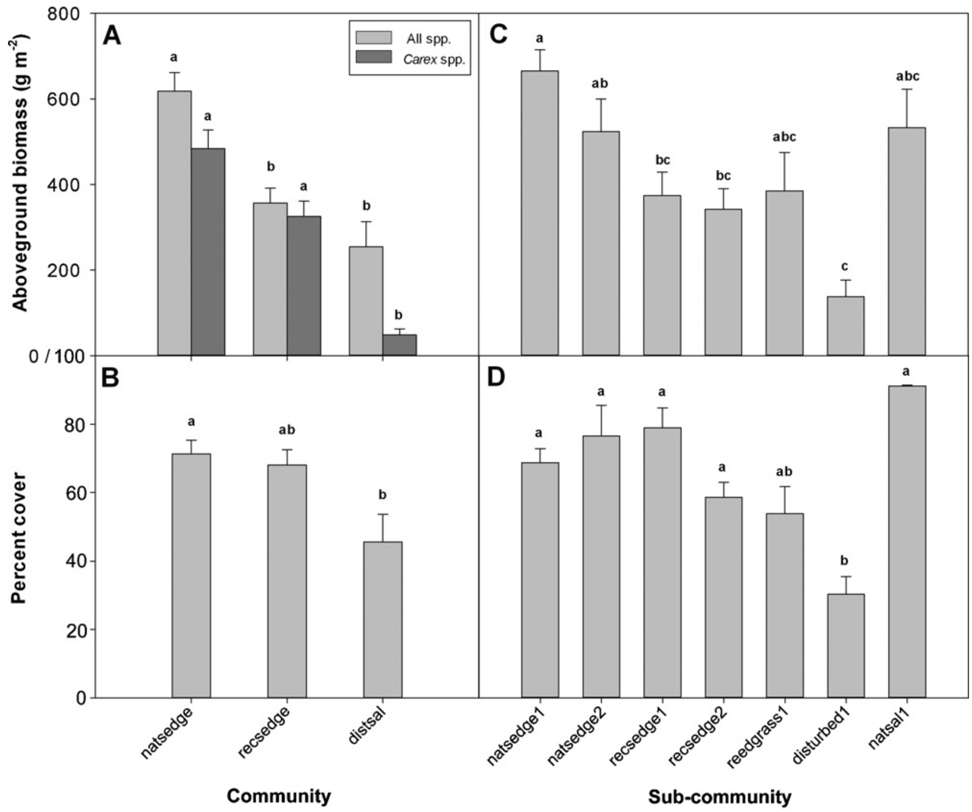

Fig. 3. (A) Wet meadow aboveground biomass by community, and *Carex* spp. contribution to aboveground biomass by community; (B) wet meadow percent cover by community; (C) wet meadow aboveground biomass by sub-community; (D) wet meadow percent cover by sub-community. Communities: natsedge = natural sedge, recsedge = reclaimed sedge, distsal = disturbed/saline. Sub-communities: natsedge1 = natural sedge I, natsedge2 = natural sedge II, recsedge1 = reclaimed sedge I, recsedge2 = reclaimed sedge II, reedgrass1= Reedgrass, disturbed1 = disturbed, natsal1 = natural saline. Bars of each series with different lower case letters are significantly different ($p < 0.01$). Vertical T-bars correspond to standard errors of the mean.

Physicochemical joint plot vectors varied along Axis 1 only (Fig. 4 inset B), and thus reflected environmental differences among the natsedge community and both the recsedge and distsal communities combined. Sites that were to the left on Axis 1 tended to have high sediment nutrients (TN and TP), and sediment water content (WM % water,) while those that were on the right of Axis 1 had higher stress scores, and NAs in the water. No physicochemical differences were detected along Axis 2.

When looking at the arrangement of the seven sub-communities, wetlands with the two natural sedge sub-communities (natsedge1 and natsedge2) overlapped within the natsedge group (Fig. 4). The remaining five sub-communities were more distinctly separated in species space, with the exception of sites with the reedgrass1sub-community, which were widely separated along Axis 1. Wetlands with the natsal1 sub-community were grouped low on Axis 2, clearly separated from other members of the distsal community and all other communities.

4. Discussion

We demonstrate the existence of three wet meadow vegetation communities among our 45 marsh wetlands. Within these three communities we show a further seven sub-communities that aid in interpreting the relationship among wetlands that had either highly variable vegetation communities in reclamation areas, or natural sites with regionally rare halophytic vegetation. This represents the largest effort yet to characterize oil sands marsh wet meadow communities, and to compare their vegetation communities with those of naturally occurring analogs of the expected reclamation outcomes. Recent experimental work has highlighted species that are tolerant to the levels of contamination present in the oil sands reclaimed wet landscape (e.g., Pouliot et al., 2012; Rezanezhad et al., 2012; Vitt et al., 2011), and our results demonstrate that some of these species are in fact already established at reclaimed wetlands, reaching levels of biomass production and vegetation cover in the range of that found at natural reference wetlands. There still exist differences in vegetation community and ecological performance among natural and reclaimed wetlands, the potential causes of which must be better understood in order to more effectively reclaim wetlands in the post-mining landscape.

4.1. Treatment type does not predict vegetation community

Construction history and constituent materials of oil sands reclaimed wetlands, which we used to assign OSREF and OSPA treatment types, were poor predictors of the vegetation communities that developed during the reclamation process. Among the reclaimed sedge (recsedge) and disturbed/saline (distsal) communities that were present at the majority of reclaimed wetlands, OSPA and OSREF sites were distributed inconsistently. This suggests that factors other than construction history and materials may be responsible for determining the vegetation communities of oil sands wetlands. Incomplete histories of the reclaimed sites limit the ability to make such conclusions definitively, however.

The majority of REF sites clustered together by sharing the natural sedge (natsedge) vegetation community that was characterized by *C. atherodes*. This community was consistent with natural vegetation identified in previous wetland inventory work

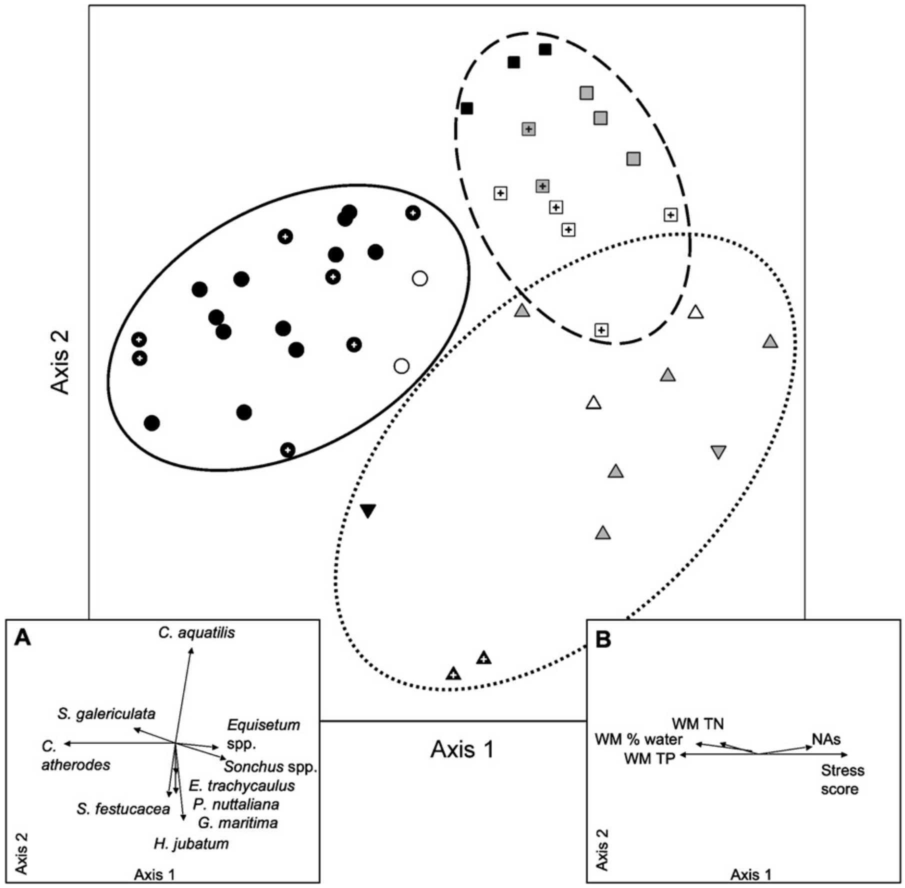

Fig. 4. Wet meadow vegetation communities visualized in species-space by NMS ordination. Site treatments are identified by color: REF (black fill), OSREF (white fill), and OSPA (gray fill). Ellipses indicate the 3 community groups identified by cluster analysis: natural sedge (solid line), reclaimed sedge (dashed line), and disturbed/saline (dotted line). The 7 sub-communities are indicated by symbols: natural sedge I (circle), natural sedge II (crossed circle), reclaimed sedge I (square), reclaimed sedge II (crossed square), disturbed (upward triangle), Reedgrass (downward triangle) and natural saline (crossed upward triangle). (A) Inset of species vector joint plot. (B) Inset of physicochemical vector joint plot. Overlay vectors are included at Pearson's correlation coefficient of $R^2 > 0.20$, and length of vectors are proportional to R^2 values.

in the boreal region of Alberta (Bayley and Mewhort, 2004; Trites and Bayley, 2009a). With the exception of two OSREF sites, the natsedge vegetation community was found exclusively at natural REF sites. OSREF sites with the natsedge community consisted of one site that was relatively undisturbed except for hydrologic alteration (SWSS Beaver Wetland), although it was found adjacent to a large tailings facility, and a second site constructed to study effectiveness of waterfowl habitat creation (Shallow Wetland). Although there were two sub-communities identified within the natsedge community (Natural sedge I [natsedge1], Natural sedge II [natsedge2]), their overlapping position in species space on the NMS ordination plot indicates it is unlikely that this division is biologically meaningful at the community level.

The recsedge community was the most tightly clustered group in ordination space, implying a high degree of similarity in vegetation community composition among member sites, and less within-group variation than was found in the other two communities. This community was rarely encountered in natural wetlands, with only three of our 25 REF sites supporting a *C. aquatilis*-dominated wet meadow (Child Lake sites 4C and 4D, Hay River site 3). Recent work using a multimetric approach to evaluate marsh wetland health among boreal and oil sands reclaimed marshes identified these three REF wetlands to be in the top 20% of best performing sites (Raab and Bayley, 2012). The first sub-community of the recsedge community, reclaimed sedge I (recsedge1), was found only at either REF or OSPA sites and identified *C. aquatilis* as the significant indicator species, which further suggests that treatment type does not determine the vegetation community. A third sedge species, *Carex utriculata* Boott, was a significant indicator of the second recsedge sub-community, Reclaimed sedge II (recsedge2), and was primarily found at OSREF sites. It is unlikely there is meaningful biological difference among the recsedge1 and recsedge2 sub-communities due to their relatively close clustering in species space.

The distsal community had the most variable vegetation of the three communities, and was the least closely grouped in species space. As there are multiple ways for oil sands reclaimed sites to become disturbed or to maintain a degraded state (e.g., disturbance type and severity, reclamation practices, hydrologic and edaphic variability), this diversity of factors can contribute to the high biological variability we observed in the distsal community (Keleher and Rader, 2008). The inclusion of two naturally saline REF sites within the distsal community highlights a problematic aspect of using only three community groups to coarsely describe all the wetland vegetation encountered in this study. The distsal community is best understood when its three constituent sub-communities (Reedgrass [reedgrass1], disturbed [disturbed1], natural saline [natsal1]) are considered separately. The two natural saline sites cluster separately from other disturbed wetlands, and contain regionally rare halophytic species found in previous surveys of naturally saline vegetation in the boreal plain (e.g., Purdy et al., 2005; Trites and Bayley, 2009a). While the two natural saline sites represent a native vegetation community adapted to higher salinities (Purdy et al., 2005), we did not encounter it on oil sands leases despite the saline water at certain reclaimed sites (Rooney et al., 2011).

The poor ability of a priori categorization of oil sands wetlands to predict vegetation communities was likely due to the diversity of reclamation materials, differing proximity of wetlands to seed sources, and variable hydrological settings which include receipt of freshwater precipitation, groundwater, or tailings pond seepage water. Individual site history is difficult to determine, beyond the broad OSPA or OSREF categorization, and sites may have been subject to undocumented additions of sediment amendments, revegetation, or specific design criteria at some point since creation. In the absence of a clear pattern of occurrence for vegetation community type on the basis of wetland construction history, there may be opportunity to intervene early and affect vegetation development by determining the best performing community to target for reclamation, as is common practice for reclaiming uplands in the oil sands (Mackenzie and Naeth, 2010).

4.2. Ecological performance of reclaimed wetlands

We identified that certain wetland vegetation communities ecologically outperformed others with respect to percent cover and aboveground biomass production. Reclaimed wetlands where *C. aquatilis* (recsedge community) and to a lesser extent *C. stricta* (reedgrass1 sub-community) have become established perform equivalently to natural wetlands with respect to vegetation percent cover, but not biomass production. Percent cover has typically been used as a measure of productivity in reclaimed wetlands (e.g., Aronson and Galatowitsch, 2008; Seabloom and van der Valk, 2003), yet we found that greater vegetation cover did not necessarily indicate that a community would have greater aboveground biomass. Our measure of August peak aboveground biomass demonstrated significantly higher production in the natsedge than the recsedge or distsal communities. When considering only *Carex* spp., biomass, however, we did not detect a significant difference among natsedge and recsedge communities. Mollard et al. (2012) found that *C. aquatilis* exposed to oil sands contaminated water showed decreased height and leaf length compared to *C. aquatilis* in natural wetlands, however our study did not collect physiological measurements at the plant-level, and there may be other factors such as increased stem density in oil sands wetlands that are compensating to produce similar *Carex* spp. aboveground biomass at the plot level. That wetlands with the recsedge community can produce aboveground biomass of *Carex* spp. within the range of that found at wetlands with the natsedge community indicates that natural levels of biomass production are possible when vegetation appropriate to the region and local abiotic conditions become established.

4.3. Revegetation targets

If uniform establishment of particular communities is desired in oil sands reclaimed wetlands, plantings may help overcome dispersal limitation at the local level. Consistent with Purdy et al. (2005), we found that while some natural colonization has already occurred, boreal wetland species have not spread throughout the reclaimed landscape. Monotypic, eutrophic marsh wetlands with low species diversity dominated by graminoids such as *C. atherodes* and *C. aquatilis* are common in the boreal plain (Trites and Bayley, 2009b), and targeting the dominant species of our natsedge or recsedge vegetation communities for reclamation would be consistent with analogous natural communities. The early planting of appropriate species may also pre-empt the establishment of less desirable species on longer timelines (Aronson and Galatowitsch, 2008; Noon, 1996), in addition to overcoming the immediate effects of dispersal limitation in newly created wetlands (Seabloom and van der Valk, 2003).

Recent experimental and plant physiology work have found *C. atherodes*, *C. utriculata, C. aquatilis* and *C. stricta* to be tolerant of levels of contamination found in oil sands process water, and demonstrate good potential for rapid revegetation of post-mining oil sands reclaimed wetlands (Mollard et al., 2012; Pouliot et al., 2012; Rezanezhad et al., 2012). With the exception of *C. atherodes*, which was not commonly encountered on oil sands leases, *C. utriculata, C. aquatilis* and *C. stricta* can be selected for planting following reclamation. The recsedge1 and reedgrass1 sub-communities in particular showed similar percent cover and aboveground biomass to the natsedge2 sub-community, and were indicated by *C. aquatilis* and *C. stricta*, respectively. Selecting these species as priorities for targeted planting following wetland creation will increase the likelihood of rapid, successful plant community establishment.

4.4. Barriers to reclamation

If equivalent land capability is to be reached by oil sands reclaimed wetlands then not only must appropriate species for revegetation be chosen, but the factors that can limit biomass production and species establishment need to be determined. Conventional thought identifies the major challenges to oil sands wetland reclamation as elevated salinity in water and sediment from weathering of saline, sodic rock, the presence of recycled process water, hydrocarbons and NAs resulting from the extraction process, fine clay particles in the tailings, heavy metal contamination, and low initial nutrient content of sediment (Harris, 2007; Johnson and Miyanishi, 2008; Kelly et al., 2009). In addition to these abiotic factors, landscape and biological processes such as dispersal limitations and wetland aging may affect reclamation outcomes.

4.4.1. Propagule arrival

Trites and Bayley (2009a) hypothesized that the paucity of *C. atherodes* in reclaimed wet meadows was due to slow dispersal of this species in the absence of connected, flowing water or waterfowl transport to reclaimed substrates. Lougheed et al. (2008) found that plant communities in ecologically degraded wetlands in Michigan were heterogeneous at the landscape level, likely due to differential colonization by plant species. The distance of reclaimed wetlands from propagule sources can determine initial revegetation success, especially in the absence of a planting or seeding program (Aronson and Galatowitsch, 2008; Houlahan et al., 2006). The closer reclaimed wetlands are to propagule sources, such as remnant wetlands on the periphery of oil sands leases, the more likely viable vegetation propagules will arrive on reclaimed substrates from nearby sources. Dispersal may currently represent the main limitation to establishing the recsedge community.

As oil sands mines expand across the landscape to eventually occupy 1670 km^2, and as the proportion of wetland decreases in the post-mining landscape (Rooney et al., 2012), the importance of dispersal will be even greater. Wetlands that have naturally received propagules from the adjacent undisturbed landscape will become rarer, and it is likely that colonization of new wetlands will decrease with fewer available wetland plant propagules. Additionally, the lack of surrounding forest cover, which may act as a barrier to invasion, can render newly created wetlands vulnerable to colonization by non-native or weedy species (Houlahan et al., 2006).

4.4.2. Age

Age can limit biomass production, and for *Carex* spp. age affects the extent of belowground rhizomatous biomass, which can serve as a source for biomass regeneration between years. While there were large differences in wetland age among community types (average age of sites with natsedge communities was assumed to be at least 1000 years, average age of sites with recsedge and distsal communities – excluding four natural REF sites – was 16 years), overall belowground biomass growth in both oil sands reclaimed wetlands and natural fresh to sub-saline boreal marshes is known to be quite low, comprising less than 10% of total annual production (Trites and Bayley, 2009b). If age does affect biomass production, it may act through peat accumulation over long time scales, enhancing sediment nutrient availability to plant growth. However, our results indicate that viable wetland vegetation communities can be established on relatively short timelines in the post-reclamation landscape if other factors such as nutrient availability are addressed.

4.4.3. Nutrients

The finding that certain reclaimed wetlands demonstrate ecological performance similar to natural wetlands suggests that the lower total aboveground biomass of the recsedge community may be due primarily to nutrient deficiencies and lack of organic matter in the sediment. We found higher sediment total N and total P at our natural sites, as well as water content of the sediment (inversely related to bulk density). NMS Axis 1 can be seen as a gradient from high sediment organic matter to the left on the axis to mineral sediment to the right on the axis (Fig. 4B). Olde Venterink et al. (2001) also found a sediment organic matter gradient to account for greater than 50% of variation in aboveground biomass of wet meadow graminoids in Europe. Fertilization as a technique to reclaim nutrient deficient mine spoils is well established for uplands (Bradshaw, 1997), and nutrient additions to mineral dune soils in North Dakota were found to increase the overall biomass of graminoid species, as well as decrease the extent of exposed sediment (Willis, 1963). Fertilization with N, P and K increased production when water was not a limiting factor, a situation that may share characteristics with the post-reclamation landscape where mineral tailings will be used as a substrate for wetlands. Fertilization of European wet grasslands with N, P and K effectively increased aboveground biomass production, however graminoid species contributed proportionally more to overall production than other species following fertilization (Vermeer and Verhoeven, 1987). The majority of boreal plain marshes are naturally eutrophic or hypereutrophic (Trites and Bayley, 2009b), and in the boreal plain, the low diversity wetlands characterized by dominant graminoids such as *C. atherodes* and *C. aquatilis* may be the optimal marsh climax community. For equivalently productive marsh communities to develop in reclaimed oil sands wetlands, reclamation will need to address the issue of nutrient deficiency, and the naturally eutrophic state of undisturbed boreal marshes indicates that fertilization may be a useful approach.

4.4.4. Salinity and contaminants

Stress from oil sands contamination can disproportionately decrease the growth of non-*Carex* spp. (Rezanezhad et al., 2012), which may also be limiting total aboveground biomass production in recsedge wetlands. It is notable that we did not find a clear salinity gradient among vegetation communities when examining physicochemical vectors produced by our NMS ordination. With the exception of the two REF sites with the halophyte-dominated natsal1 sub-community it is not clear that salinity is affecting the vegetation communities we observed, in spite of the salinity gradient from fresh to sub-saline along which the natural REF and reclaimed OSPA and OSREF sites were distributed.

While salinity is known to affect wetland vegetation (e.g., Houle et al., 2001; Lewis and Weber, 2002; Lissner and Schierup, 1997), and shape the vegetation community toward halophyte-dominance at natural wetlands in the boreal plain (Purdy et al., 2005; Trites and Bayley, 2009a), matching vegetation to salinity may not, at least initially, be critical to reclamation success. If we measure success as the establishment of vegetation communities similar to those found naturally in shallow open water marshes in the boreal region, then actively planting species such as *C. atherodes* and *C. aquatilis* which are tolerant of the salinity levels found at oil sands reclaimed sites (Mollard et al., 2012; Rezanezhad et al., 2012; Trites and Bayley, 2009b), may supplant weedy or invasive species and contribute to substrate stabilization and organic matter accumulation.

Both the abiotic stress scores and NAs increased across NMS Axis 1 from REF wetlands toward the OSREF and OSPA wetlands (Fig. 4 inset B). We did not analyze pore water of wet meadow sediment, however NAs are a known constituent of process affected tailings materials, and their presence in the water is likely indicative of NA presence in the sediment of reclaimed wetlands. While NA presence is associated with NMS Axis 1, the relatively well performing recsedge community is separated from the poorly performing distsal vegetation community along Axis 2, and it does not seem likely that NAs are affecting species composition or ecological performance in reclaimed marshes.

5. Management implications

Oil sands reclaimed wetland vegetation communities tend toward either a sedge-dominated wet meadow community (recsedge), or an invasive species and halophyte dominated wet meadow community (distsal). On the basis of the barriers to reclamation that we have identified, and in the absence of clear physical or chemical differences among these two reclaimed communities, reclamation managers should consider that early intervention with planting of *C. aquatilis* (known to be tolerant of oil sands contaminants), fertilization to enhance sediment nutrients, and construction of wetlands adjacent to propagule sources or with some interconnectivity will best direct the development of reclaimed wetland vegetation communities toward that of natural boreal marshes. If managers are seeking to meet reclamation goals of healthy, functioning wetlands within the first decade following mine closure, strong initial efforts to establish specific vegetation communities in reclaimed wetlands may be the best method to increase success (Gutrich et al., 2009). The reclamation timeline, or speed at which successful reclamation is expected or required, will determine whether early intervention at newly constructed wetlands is critical, as short timelines will require at the very least propagule introduction at a faster rate than would occur naturally.

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Appendix A. Supplementary data

Supplementary data associated with this article can be found, in the online version, at http://dx.doi.org/10.1016/j.ecoleng.2013.01.024.

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